AMINES

MOP of Amines

① Reduction of nitro compounds

$$Ph-NO_2 \xrightarrow[\text{Sn/HCl or Fe/HCl}]{H_2+Pd/\text{Alcohol or}} Ph-NH_2$$

NO_2 hatao NH_2 lagao

★ OP Point :-

Aliphatic nitro $R-NO_2 \xrightarrow{LAH} R-NH_2$

↳ but LAH cannot give NH_2 of aromatic nitro.

② Ammonolysis of alkyl halides

$$\ddot{N}H_3 \xrightarrow[1^\circ\ S_N2]{R-X} R-\overset{+}{N}H_3 + X^{\ominus} \longrightarrow R-NH_2 + HX \quad (1^\circ \text{ Amine})$$

$$R-NH_2 \xrightarrow{R-X} R_2NH \ (2^\circ \text{ amine}) \xrightarrow{R-X} R_3\ddot{N} \ (3^\circ \text{amine}) \xrightarrow{R-X} R_4\overset{+}{N}\overset{-}{X}$$

$R_4\overset{+}{N}\overset{-}{X}$ quaternary Ammonium salt

③ Reduction of nitriles

$$R-C\equiv N \xrightarrow[\text{3) Na(Hg) + ROH}]{\text{1) LAH/}H_2O \quad \text{2) }H_2+Ni} R-CH_2-NH_2$$

CN Hatao CH_2NH_2 lagao

★ OP Point :-

↳ LAH can reduce Isocyanide

$$R-\overset{+}{N}\equiv\overset{-}{C} \xrightarrow{LAH} R-NH-CH_3$$

$\boxed{\text{NC hatao, } NHCH_3 \text{ lagao}}$

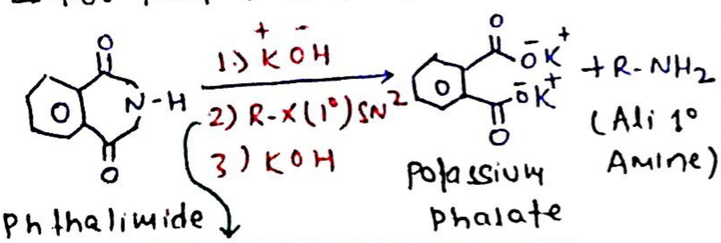
④ Reduction of Amides

$$R-\overset{O}{\overset{||}{C}}-NH_2 \xrightarrow{LAH} R-CH_2-NH_2$$

$\boxed{\text{CO hatao, } CH_2 \text{ lagao}}$

⑤ Gabriel phthalimide synthesis

↳ For prep 1° Aliphatic amine



$\boxed{\text{X hatao, } NH_2 \text{ lagao}}$

Intermidiate :- N-alkylated phthalimide

⑥ Hoffmann Bromamide degradation Rxn

$$R-\overset{O}{\overset{||}{C}}-NH_2 \xrightarrow[KOH]{Br_2} R-NH_2$$

CO ko phenk do, uda do jo krna h kro ☺

★ OP Point :-

① Nitrene intermediate $R-\overset{O}{\overset{||}{C}}-\ddot{N}:$

② Bromamide " $R-\overset{O}{\overset{||}{C}}-NH-Br$

③ Overall Rxn

$$\boxed{R-\overset{O}{\overset{||}{C}}-NH_2 + Br_2 + 4KOH \rightarrow R-NH_2 + 2KBr + K_2CO_3 + 2H_2O}$$

Properties of Amine

① Basic strength

- $B.S \propto +M, +H, +I \propto \frac{1}{-M} \propto \frac{1}{-H} \propto \frac{1}{-I}$
- In gas phase :- $RNH_2 < R_2NH < R_3N$
- In Aq medium :-
  - $R = Me \rightarrow 2^\circ > 1^\circ > 3^\circ$
  - $R = Et \rightarrow 2^\circ > 3^\circ > 1^\circ$
  - $R = Pr \rightarrow 3^\circ > 2^\circ > 1^\circ$
  - ⋮ ⋮

② Alkylation :- (Ammonolysis)

$$NH_3 + CH_3-Cl \longrightarrow \underset{1^\circ}{CH_3NH_2} \xrightarrow{CH_3Cl} \underset{2^\circ}{CH_3-NH-CH_3} \xrightarrow{CH_3-Cl} \underset{3^\circ}{CH_3-\underset{CH_3}{\underset{|}{N}}-CH_3} \xrightarrow{CH_3-Cl} Me_4\overset{+}{N}\overset{-}{Cl}$$

$Me_4\overset{+}{N}\overset{-}{Cl}$ 4° ammonium salt

③ Acylation :- (only one time)

$$R-NH_2 \xrightarrow{CH_3-\overset{O}{\overset{||}{C}}-Cl} CH_3-\overset{O}{\overset{||}{C}}-NH-R + HCl$$

$\boxed{\text{H Hatao, COR lagao}}$

④ Carbyl Amine Test

↳ Isocyanide Test for 1° Amine

$$R-NH_2 \xrightarrow[KOH]{CHCl_3} R-NC$$

NH_2 hatao NC lagao

⑤ Rxn with Nitrous Acid

ⓐ With 1° Ali amine

$$R-NH_2 \xrightarrow[\text{or } NaNO_2 + HCl]{HNO_2} R\overset{+}{N_2}\overset{-}{Cl} \longrightarrow \overset{+}{R} + N_2 + Cl^-$$

$\overset{+}{R} \xrightarrow[H_2O]{-H^+} ROH$; $\overset{+}{R}$ ↓ Rearrange, POS ; Cl^- ↓ H^+ HCl

② Aro 1° Amine

$$PhNH_2 \xrightarrow[HCl]{NaNO_2} Ph\overset{+}{N_2}Cl^-$$

③ Ali & Aro 2° Amine

$$R_2N\boxed{H + HO}NO \longrightarrow R_2N-NO$$

H hatao, NO lagao

Yellow oily nitrosamine

④ Ali 3° Amine

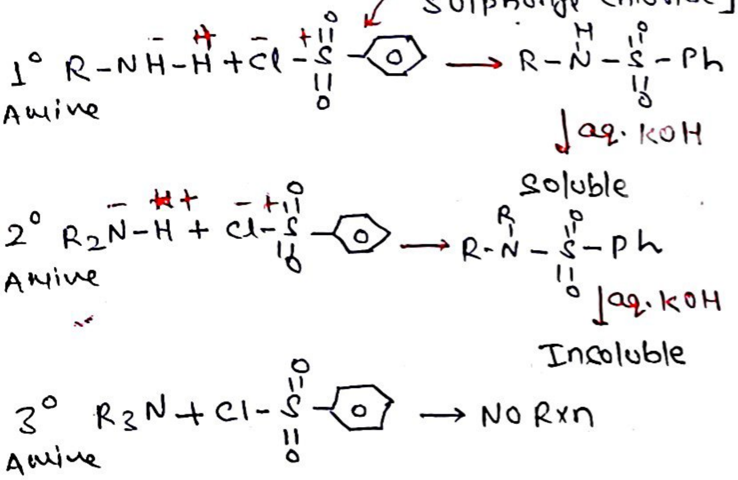
$$R_3N: \overset{\oplus}{NO} \longrightarrow R_3\overset{\oplus}{N}-NO$$

⑤ Aro 3° Amine

⑥ Hinsberg's Test

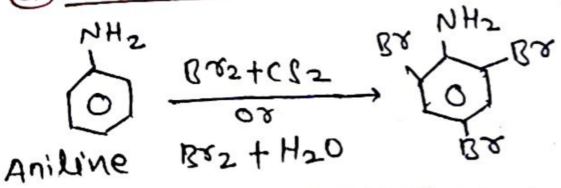
↳ For disting. b/w 1°, 2°, 3°.

Hinsberg's Reagent :- $PhSO_2Cl$ [Benzene sulphonyl chloride]

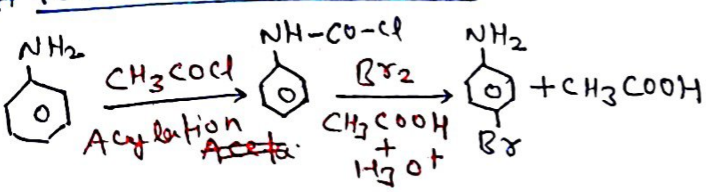


⑦ E.S.R

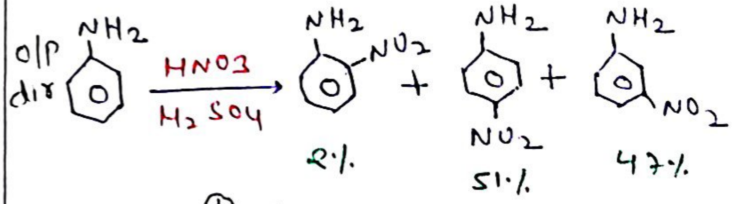
ⓐ Bromination

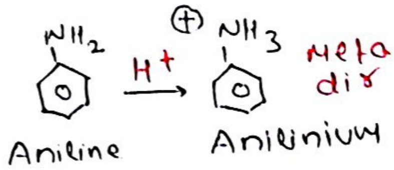


For Mono-bromination

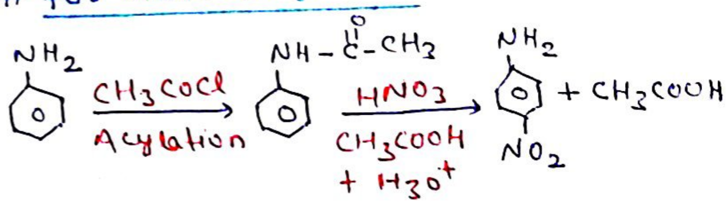


ⓑ Nitration

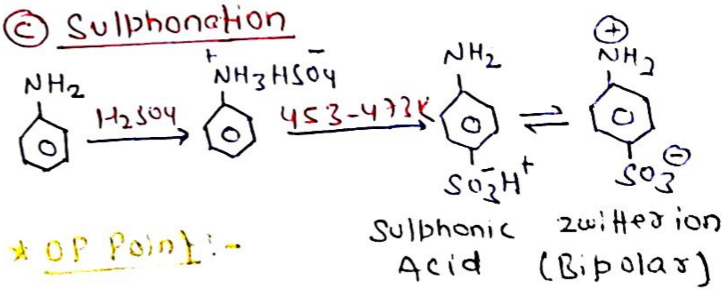




For Mono-Nitration



ⓒ Sulphonation



★ OP Point :-

↳ Aniline doesn't give f.c.R

MOP of diazonium salt

$$Ph-NH_2 \xrightarrow[0-5°C]{NaNO_2+HCl} Ph\overset{+}{N_2}Cl^-$$

Properties of diazonium salt